Name- Surname:

Student Number:

05.02.2021

FINAL EXAM

If your student number ends with

(1) even number except '0'

Answer Q1, Q4, Q6

(2) odd number except '5'

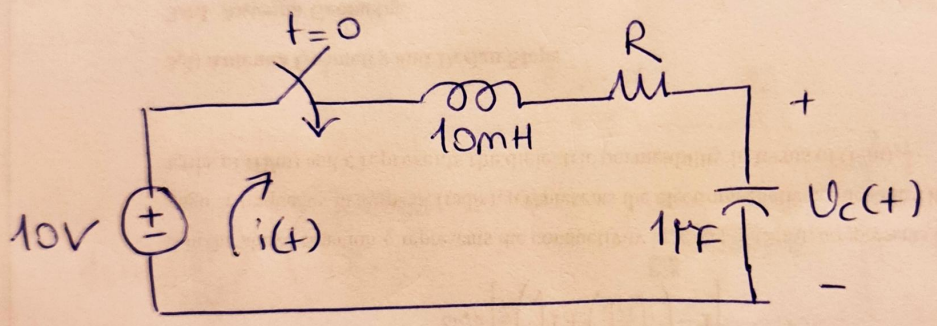
Answer Q2, Q5, Q7

(3) '0' OR '5'

Answer Q3, Q4, Q7

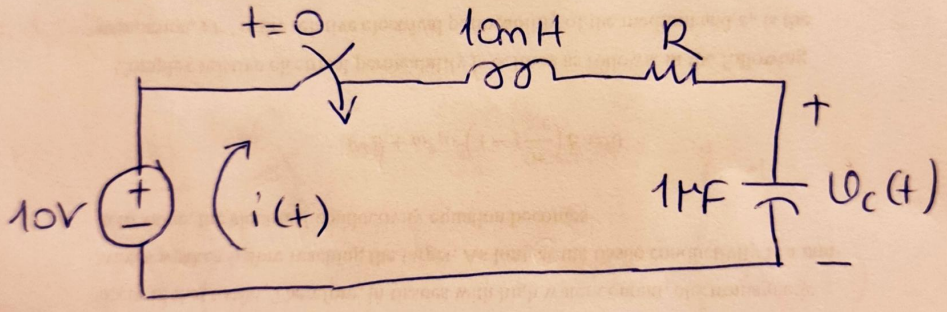
Q1

A dc source is connected to a series RLC circuit by a switch that closes at $t=0$ as show in the figure. The initial conditions are $i(0)=0$ and $v_c(0)=0$. Solve for $v_c(t)$ if $R=300\,\Omega$



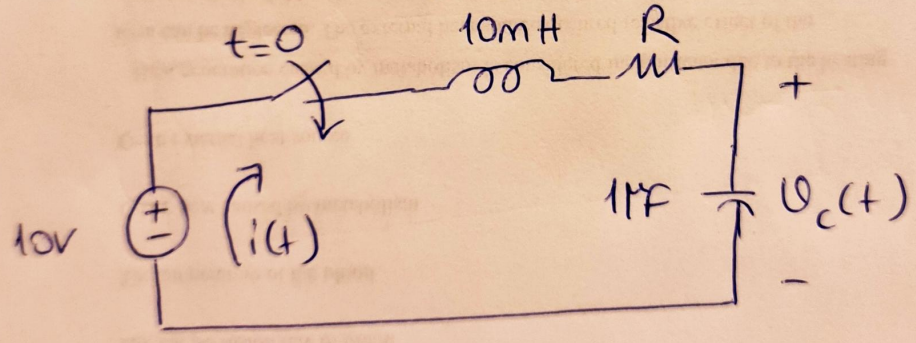
Q2

A dc source is connected to a series RLC circuit by a switch that closes at $t=0$ as shown in the figure. The initial conditions are $i(0)=0$ and $v_c(0)=0$. Solve for $v_c(t)$ if $R=200\Omega$.



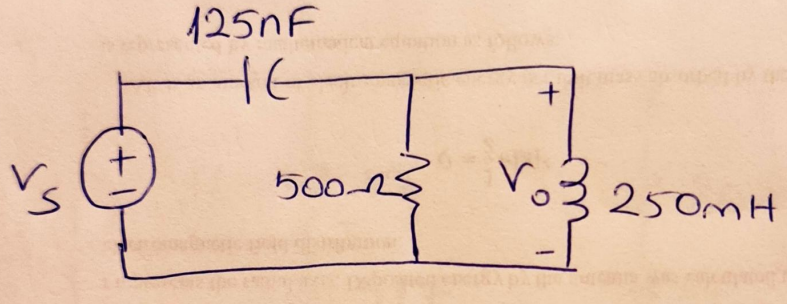
Q3

A dc source is connected to a series RLC circuit by a switch that closes at $t=0$ as shown in the figure. The initial conditions are $i(0)=0$ and $v_c(0)=0$. Solve for $v_c(t)$ if $R=100\,\Omega$.



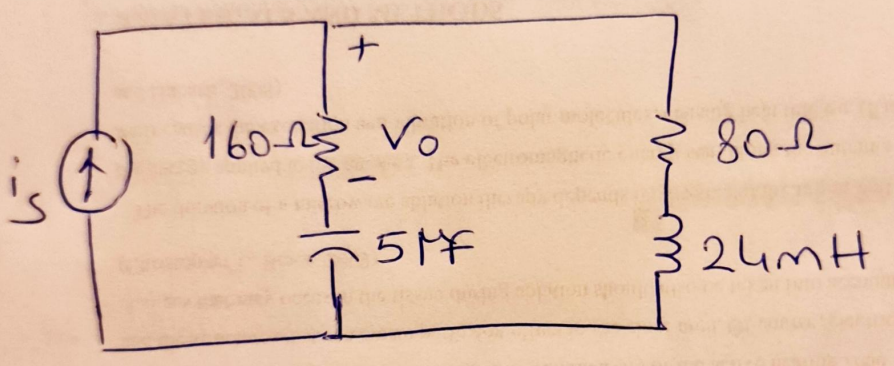
Q4

The circuit in the figure is operating in the sinusoidal steady state. Find the steady-state expression for $v_o(t)$ if $v_s = 32\cos 4000t$ V.



Q5

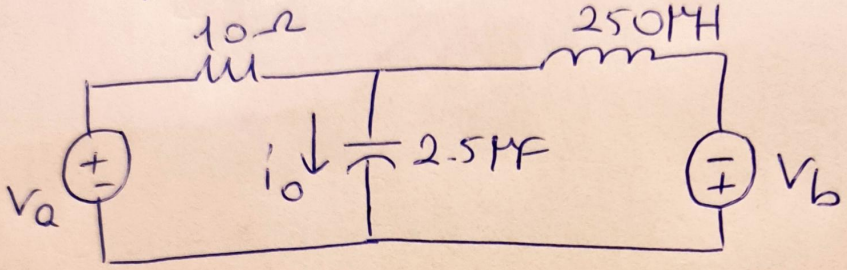
Find the steady-state expression for v_o in the circuit of the figure if $i_s = 150\cos 200t$ mA.



Q6

Use the mesh-current method to find the steady state expression for $i_o(t)$ in the circuit if $v_a = 30\cos 20000t$ V,

$v_b = 60\sin(20000t + 60°)$ V.



Q7

Use the node-voltage method to find the phasor voltage V_g in the circuit.

